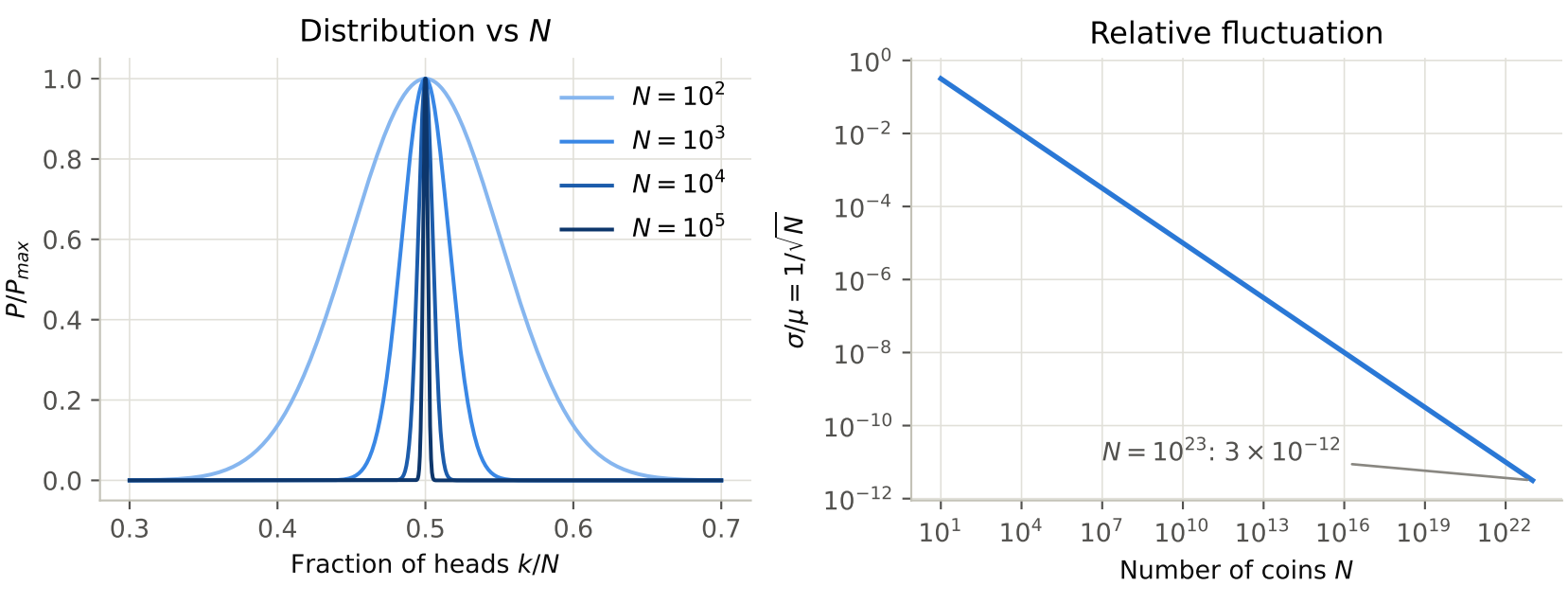
Distribution vs N
$N = 10^2$
$N = 10^3$
$N = 10^4$
$N = 10^5$
P/P_{max}
1.0
0.8
0.6
0.4
0.2
0.0
0.3
0.4
0.5
0.6
0.7
Fraction of heads k/N
Relative fluctuation
$\sigma/\mu = 1/\sqrt{N}$
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-12}
$N = 10^{23}$: 3×10^{-12}
10^{1}
10^{4}
10^{7}
10^{10}
10^{13}
10^{16}
10^{19}
10^{22}
Number of coins N